

Why WiMAX Will Not Fail

WiMAX provides robust access control, data privacy and data integrity using sophisticated authentication and encryption technologies. This article investigates the architecture enabling the high-level security

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In today's fast-paced lifestyle, consumers want communication at faster speed and lower cost, more broadband capabilities as well as nomadic and mobility support. Keeping these demands in mind, IEEE 802.16 working group has come up with a new wireless communications standard called the broadband wireless access (BWA). Commonly known as WiMAX (worldwide interoperability for microwave access), this is a fast evolving technology used to form wide-range wireless networks with high data rate of information transfer.

We live in a variety of networking environments for using Internet-based services and applications. Data security becomes an important issue in such interconnected networks in order to safely transmit or receive information. Due to this, different security protocols are designed and deployed with network standards.

The latest update given by WiMAX Forum—the IEEE 802.16m—claims to offer up to 100Mbps mobile and 1Gbps fixed speeds. The update is added to the WiMAX standard in both its versions—fixed (IEEE 802.16d-2004) and mobile (IEEE 802.16e-2005) broadband wireless access.

WiMAX protocol

IEEE 802.16 standard essentially specified two aspects of the air interface for WiMAX—the physical layer (PHY) and the media access control layer (MAC) (refer Fig. 1). The physical layer defines electrical and physical specifications for devices, establishment and termination of a connection to a communication medium, communication flow

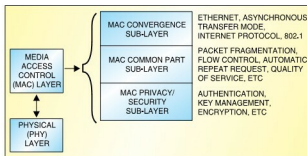


Fig. 1: WiMAX protocol layers

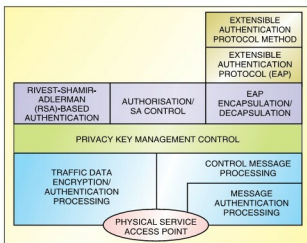


Fig. 2: WiMAX security sublayer

control, modulation, coding, etc. On the other hand, MAC layer is further divided into three sub-layers—convergence, common part, and security or privacy.

The convergence sub-layer describes how wireline technologies such as asynchronous transfer mode (ATM), Ethernet, 802.1 (LAN/MAN) and Internet protocol (IP) are encapsulated on the air interface and how data is classified. The common part sub-layer is responsible for idle-mode processes like cell selection, paging structures and location-area updates. This layer is also responsible for sleep-mode

processes, handover procedures, multicast and broadcast services, quality-of-service (QoS) class and automatic repeat request (ARQ) processes. It also does header suppression, packing and fragmentation for efficient use of spectrum.

The security sub-layer provides subscribers with privacy, authentication or confidentiality across the broadband wireless network. It is accomplished by applying cryptographic transforms to MAC packet data units carried across connections between the subscriber station and the base station. Secure communications are delivered

by using secure key exchange during authentication, and encryption using advanced encryption standard (AES) or data encryption standard (DES) during data transfer. The MAC layer incorporates privacy key management version 2 (PKMv2) for MAC layer security. PKMv2 incorporates support for extensible authentication protocol (EAP).

In addition, the security sub-layer provides operators strong protection from theft of service. The base station protects against unauthorised access to data transport services by securing the associated service flows across the net-